

Arbitrary Decimation of a Gaussian Army

Deconvolution, Residual Distributions, and Applications in Statistics, Signal Processing, and Machine Learning

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Abstract

Given two Gaussian probability density functions $m \sim \mathcal{N}(\mu, \sigma^2)$ and $n \sim \mathcal{N}(\nu, \tau^2)$, we characterise the unique density d whose convolution with m equals n . We call this operation *Gaussian decimation*: the systematic extraction of one Gaussian component from a composite Gaussian. The central result states that $d \sim \mathcal{N}(\nu - \mu, \tau^2 - \sigma^2)$ whenever $\tau^2 > \sigma^2$, that d degenerates to a Dirac delta when $\tau^2 = \sigma^2$ and $\nu = \mu$, and that no real Gaussian solution exists when $\tau^2 < \sigma^2$. We prove the result via characteristic functions, analyse all three variance regimes, extend the theorem to multivariate Gaussians, provide a numerically robust algorithm, and draw connections to Bayesian inference, Kalman filtering, Gaussian process regression, variational autoencoders, and score-based diffusion models.

The paper ends with “The End”

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1 Introduction

The Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$ is the cornerstone of classical statistics, signal processing, and modern machine learning. Its closure under convolution—the sum of independent Gaussians is again Gaussian—makes it uniquely tractable for analytic computation. Yet this same stability raises a natural inverse question:

Given that the convolution of an unknown density d with a known Gaussian m equals a second known Gaussian n , what is d ?

We call the process of identifying and removing a Gaussian component **Gaussian decimation**, by analogy with the Roman military practice of reducing a unit by one tenth. Here the “army” is the collection of Gaussian components whose aggregate convolution forms n ; decimation singles out one member (m) and determines the remainder (d).

The problem arises throughout applied mathematics and data science.

- In **signal processing**, a received signal is the convolution of a transmitted source with a channel impulse response; Gaussian decimation deconvolves the channel.
- In **Bayesian inference**, the marginal likelihood is the convolution of the prior with the likelihood; inverting this yields either factor given the other.
- In **machine learning**, Gaussian process priors, variational autoencoders, and score-based diffusion models all manipulate Gaussian convolutions implicitly, and decimation illuminates the structure of each.
- In **Kalman filtering**, the prediction step convolves the filtered state density with the process-noise kernel; Gaussian decimation inverts that step, enabling system identification.

The remainder of the paper is organised as follows. Section 2 establishes notation and prerequisites. Section 3 states and proves the main theorem. Section 4 analyses all three variance regimes. Section 5 extends the result to multivariate Gaussians. Section 6 provides a numerically robust algorithm. Section 7 surveys classical applications. Section 8 draws connections to AI and machine learning. Section 9 illustrates the theorem geometrically. Section 10 concludes.

2 Preliminaries

2.1 Notation and Basic Definitions

We work throughout on the real line $\mathbb{R}$ equipped with the Borel σ -algebra and Lebesgue measure.

Definition 2.1 (Gaussian probability density). For $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, the probability density function (pdf) of the normal distribution $\mathcal{N}(\mu, \sigma^2)$ is

$$\phi_{\mu, \sigma^2}(x) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Definition 2.2 (Convolution of densities). For Lebesgue-integrable functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$, their convolution is

$$(f * g)(x) := \int_{-\infty}^{\infty} f(x-t) g(t) dt, \quad x \in \mathbb{R}.$$

If $X \sim f$ and $Y \sim g$ are independent random variables, then $X + Y \sim f * g$.

Definition 2.3 (Characteristic function). The *characteristic function* of a random variable X with density f is

$$\hat{f}(\xi) := \mathbb{E}[e^{i\xi X}] = \int_{-\infty}^{\infty} e^{i\xi x} f(x) dx, \quad \xi \in \mathbb{R}.$$

For $X \sim \mathcal{N}(\mu, \sigma^2)$, a standard Gaussian integral yields

$$\hat{f}(\xi) = \exp(i\mu\xi - \tfrac{1}{2}\sigma^2\xi^2). \tag{1}$$

The key analytic tool is the classical convolution–multiplication duality.

Lemma 2.4 (Convolution–multiplication duality). *For Lebesgue-integrable densities f and g ,*

$$\widehat{f * g}(\xi) = \widehat{f}(\xi) \cdot \widehat{g}(\xi), \quad \xi \in \mathbb{R}.$$

Proof. Fubini’s theorem applies because f and g are integrable. Setting $u = x - t$ and interchanging the order of integration:

$$\begin{aligned} \widehat{f * g}(\xi) &= \int_{-\infty}^{\infty} e^{i\xi x} \int_{-\infty}^{\infty} f(x - t) g(t) dt dx \\ &= \int_{-\infty}^{\infty} g(t) e^{i\xi t} \int_{-\infty}^{\infty} f(u) e^{i\xi u} du dt = \widehat{f}(\xi) \cdot \widehat{g}(\xi). \end{aligned} \quad \square$$

2.2 The Decimation Problem

Definition 2.5 (Gaussian decimation problem). Let $m = \phi_{\mu, \sigma^2}$ and $n = \phi_{\nu, \tau^2}$ be Gaussian densities. The *Gaussian decimation problem* is to find a probability density function d on $\mathbb{R}$ such that

$$m * d = n.$$

Equivalently, if $X \sim m$ is a known source of randomness, we seek the law of an independent “residual” random variable Z such that $X + Z \sim n$.

3 Main Result

Theorem 3.1 (Gaussian Decimation). *Let $m = \phi_{\mu, \sigma^2}$ and $n = \phi_{\nu, \tau^2}$ with $\tau^2 > \sigma^2$. The unique solution to the Gaussian decimation problem within the class of Gaussian densities is*

$$\boxed{d = \phi_{\nu - \mu, \tau^2 - \sigma^2}} = \frac{1}{\sqrt{2\pi(\tau^2 - \sigma^2)}} \exp\left(-\frac{(x - (\nu - \mu))^2}{2(\tau^2 - \sigma^2)}\right), \quad (2)$$

i.e., d is the pdf of $\mathcal{N}(\nu - \mu, \tau^2 - \sigma^2)$.

Proof. Step 1: Reduce to characteristic functions. By Lemma 2.4, the equation $m * d = n$ is equivalent to the pointwise identity

$$\widehat{m}(\xi) \cdot \widehat{d}(\xi) = \widehat{n}(\xi), \quad \forall \xi \in \mathbb{R}.$$

Step 2: Solve by pointwise division. The Gaussian characteristic function (1) satisfies $\widehat{m}(\xi) = \exp(i\mu\xi - \frac{1}{2}\sigma^2\xi^2) \neq 0$ for every $\xi \in \mathbb{R}$, so we may divide:

$$\begin{aligned} \widehat{d}(\xi) &= \frac{\widehat{n}(\xi)}{\widehat{m}(\xi)} = \frac{\exp(i\nu\xi - \frac{1}{2}\tau^2\xi^2)}{\exp(i\mu\xi - \frac{1}{2}\sigma^2\xi^2)} \\ &= \exp(i(\nu - \mu)\xi - \frac{1}{2}(\tau^2 - \sigma^2)\xi^2). \end{aligned} \quad (3)$$

Step 3: Identify the distribution. Expression (3) is precisely the Gaussian characteristic function (1) with mean $\nu - \mu$ and variance $\tau^2 - \sigma^2 > 0$. By the *Lévy uniqueness theorem*, the characteristic function determines the distribution uniquely, so d is the Gaussian density $\phi_{\nu - \mu, \tau^2 - \sigma^2}$.

Step 4: Direct verification.

$$\begin{aligned} \widehat{m}(\xi) \cdot \widehat{d}(\xi) &= \exp(i\mu\xi - \frac{1}{2}\sigma^2\xi^2) \cdot \exp(i(\nu - \mu)\xi - \frac{1}{2}(\tau^2 - \sigma^2)\xi^2) \\ &= \exp(i\nu\xi - \frac{1}{2}\tau^2\xi^2) = \widehat{n}(\xi). \end{aligned} \quad \square$$

Corollary 3.2 (Probabilistic interpretation). *If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Z \sim \mathcal{N}(\nu - \mu, \tau^2 - \sigma^2)$ are independent, then $X + Z \sim \mathcal{N}(\nu, \tau^2)$.*

Proof. Immediate from Theorem 3.1: $X + Z$ has density $m * d = n = \phi_{\nu, \tau^2}$. $\square$

Remark 3.3. Theorem 3.1 is the converse of the classical *Gaussian stability* property. Stability asserts that adding two independent Gaussians yields a Gaussian with means and variances that add separately. Decimation asserts that the complementary factor is also Gaussian, provided the target distribution is at least as spread out as the component being removed.

4 Conditions and Degenerate Cases

4.1 The Three Regimes

The feasibility of Gaussian decimation depends entirely on the sign of $\tau^2 - \sigma^2$. Table 1 summarises all cases.

Table 1: Regimes of the Gaussian decimation problem $m * d = n$, with $m \sim \mathcal{N}(\mu, \sigma^2)$ and $n \sim \mathcal{N}(\nu, \tau^2)$.

Condition	Mean of d	Variance of d	Outcome
$\tau^2 > \sigma^2$ (generic)	$\nu - \mu$	$\tau^2 - \sigma^2 > 0$	Unique Gaussian pdf
$\tau^2 = \sigma^2, \nu = \mu$	0	0	Dirac delta $\delta(x)$
$\tau^2 = \sigma^2, \nu \neq \mu$	—	—	No valid pdf exists
$\tau^2 < \sigma^2$ (any ν)	—	< 0	No real Gaussian pdf

4.2 The Dirac Limit

As $\tau^2 \downarrow \sigma^2$ and simultaneously $\nu \rightarrow \mu$, the decimating density concentrates at the origin:

$$\phi_{\nu-\mu, \tau^2-\sigma^2}(x) \xrightarrow{\tau^2 \rightarrow \sigma^2, \nu \rightarrow \mu} \delta(x),$$

consistently with the convolution identity $m * \delta = m$.

4.3 Impossibility When $\tau^2 < \sigma^2$

Convolution of non-degenerate densities strictly increases variance. For independent random variables X and Z with finite second moments,

$$\text{Var}(X + Z) = \text{Var}(X) + \text{Var}(Z).$$

Consequently, $m * d = n$ requires

$$\tau^2 = \sigma^2 + \text{Var}(d) \geq \sigma^2,$$

with equality only when $\text{Var}(d) = 0$, i.e., when $d = \delta(x)$ is a point mass. No genuine positive-variance density d can reduce the spread of m through convolution; in military terms, the army can only grow, never shrink.

5 Generalisation to Multivariate Gaussians

The result extends seamlessly to p dimensions by replacing the scalar variance condition with positive-definiteness of the residual covariance.

Theorem 5.1 (Multivariate Gaussian Decimation). *Let $m \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $n \sim \mathcal{N}(\boldsymbol{\nu}, \mathbf{T})$ be p -dimensional Gaussian densities. If $\mathbf{T} - \boldsymbol{\Sigma}$ is symmetric positive definite, the unique Gaussian solution to $m * d = n$ is*

$$d \sim \mathcal{N}(\boldsymbol{\nu} - \boldsymbol{\mu}, \mathbf{T} - \boldsymbol{\Sigma}).$$

Proof. The multivariate characteristic function of $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ at $\boldsymbol{\xi} \in \mathbb{R}^p$ is

$$\widehat{m}(\boldsymbol{\xi}) = \exp\left(i\boldsymbol{\xi}^\top \boldsymbol{\mu} - \frac{1}{2}\boldsymbol{\xi}^\top \boldsymbol{\Sigma} \boldsymbol{\xi}\right),$$

which is everywhere non-zero. Pointwise division gives

$$\widehat{d}(\boldsymbol{\xi}) = \frac{\widehat{n}(\boldsymbol{\xi})}{\widehat{m}(\boldsymbol{\xi})} = \exp\left(i\boldsymbol{\xi}^\top (\boldsymbol{\nu} - \boldsymbol{\mu}) - \frac{1}{2}\boldsymbol{\xi}^\top (\mathbf{T} - \boldsymbol{\Sigma}) \boldsymbol{\xi}\right).$$

This is the characteristic function of a valid Gaussian distribution if and only if $\mathbf{T} - \boldsymbol{\Sigma} \succ 0$. The result then follows from the Lévy uniqueness theorem. $\square$

Table 2 places the univariate and multivariate settings side by side.

Table 2: Univariate and multivariate Gaussian decimation compared.

Property	Univariate	Multivariate (p dimensions)
Mean of d	$\nu - \mu$	$\boldsymbol{\nu} - \boldsymbol{\mu}$
Spread of d	$\tau^2 - \sigma^2$	$\mathbf{T} - \boldsymbol{\Sigma}$
Existence condition	$\tau^2 > \sigma^2$	$\mathbf{T} - \boldsymbol{\Sigma} \succ 0$
Degenerate case	$\tau^2 = \sigma^2, \nu = \mu$	$\mathbf{T} = \boldsymbol{\Sigma}, \boldsymbol{\nu} = \boldsymbol{\mu}$
Key analytic tool	Scalar Fourier transform	Vector characteristic function

6 Algorithmic Procedure

6.1 Exact Algorithm

Algorithm 1 implements Gaussian decimation with explicit feasibility checks.

Algorithm 1 Gaussian Decimation

Require: Parameters (μ, σ^2) of m ; parameters (ν, τ^2) of n ; numerical tolerance $\varepsilon > 0$.

Ensure: Parameters (μ_d, σ_d^2) of the decimating density d , or a diagnostic code.

```

1:  $\delta_\mu \leftarrow \nu - \mu$ 
2:  $\delta_v \leftarrow \tau^2 - \sigma^2$ 
3: if  $\delta_v > \varepsilon$  then
4:    $\mu_d \leftarrow \delta_\mu$ 
5:    $\sigma_d^2 \leftarrow \delta_v$ 
6:   return  $(\mu_d, \sigma_d^2)$   $\triangleright d \sim \mathcal{N}(\mu_d, \sigma_d^2)$ 
7: else if  $|\delta_v| \leq \varepsilon$  and  $|\delta_\mu| \leq \varepsilon$  then
8:   return DIRAC(0)  $\triangleright d = \delta(x)$ ; the two distributions coincide
9: else
10:  return NoSolution  $\triangleright \tau^2 < \sigma^2$ , or means differ with equal variances
11: end if
```

6.2 Numerical Considerations

Catastrophic cancellation can occur in floating-point arithmetic when $\tau^2 \approx \sigma^2$, because the subtraction $\tau^2 - \sigma^2$ amplifies relative rounding errors. Three mitigations are advisable.

1. **Log-space representation.** Store $\log \sigma^2$ and $\log \tau^2$ when variances span many orders of magnitude; compute $\delta_v = \tau^2 - \sigma^2$ only as a final exponentiation step.
2. **Relative-difference threshold.** Accept the result of the subtraction as numerically reliable only when $(\tau^2 - \sigma^2)/\tau^2 > \varepsilon_{\text{machine}}^{1/2}$.
3. **Graceful degradation.** When the threshold is not met, return the Dirac-delta code rather than a density with a spuriously small—or negative—variance.

7 Applications in Statistics and Signal Processing

7.1 Channel Equalisation

A Gaussian source $X \sim \mathcal{N}(\mu, \sigma^2)$ is transmitted over an additive white Gaussian noise (AWGN) channel with noise $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$, yielding a received signal $Y = X + \varepsilon \sim \mathcal{N}(\mu, \sigma^2 + \sigma_n^2)$. Knowledge of the transmitted and received marginals immediately recovers the noise distribution via Gaussian decimation:

$$\varepsilon \sim \mathcal{N}(0, \text{Var}(Y) - \text{Var}(X)) = \mathcal{N}(0, \sigma_n^2).$$

This identity underlies minimum-mean-square-error (MMSE) estimation of X from Y and provides a foundation for optimal linear equalisers.

7.2 Bayesian Deconvolution

Suppose a parameter has prior $\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$ and the marginal distribution of an observation is $Y \sim \mathcal{N}(\mu_0 + \mu_\varepsilon, \sigma_0^2 + \sigma_\varepsilon^2)$. Because $Y = \theta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(\mu_\varepsilon, \sigma_\varepsilon^2)$ independent of θ , the marginal decomposes as $p(y) = p(y | \theta) * p(\theta)$. Gaussian decimation then identifies the observation-noise density:

$$p(y | \theta) = \phi_{\mu_\varepsilon, \sigma_\varepsilon^2}(y - \theta).$$

7.3 Kalman Filter: Inverting the Prediction Step

In a linear Gaussian state-space model the Kalman prediction step gives

$$p(\mathbf{x}_t | \mathbf{y}_{1:t-1}) = \int p(\mathbf{x}_t | \mathbf{x}_{t-1}) p(\mathbf{x}_{t-1} | \mathbf{y}_{1:t-1}) d\mathbf{x}_{t-1},$$

which is a Gaussian convolution [5]. Given the filtered density at step $t-1$ and the predicted density at step t , Theorem 5.1 recovers the transition density $p(\mathbf{x}_t | \mathbf{x}_{t-1})$ directly—enabling *system identification* from observed state sequences without access to the transition model.

8 Connections to Machine Learning and AI

8.1 Variational Autoencoders

A variational autoencoder (VAE) [6] places a standard Gaussian prior $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ on the latent code and trains a decoder $p_\theta(\mathbf{x} | \mathbf{z})$. When the decoder is Gaussian, the marginal $p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x} | \mathbf{z}) p(\mathbf{z}) d\mathbf{z}$ is a Gaussian convolution, and Gaussian decimation characterises the decoder covariance from the encoder statistics and the learned marginal.

8.2 Score-Based Diffusion Models

Denoising diffusion probabilistic models [4] define a forward Markov chain that corrupts data $\mathbf{x}_0 \sim p_{\text{data}}$ by stepwise addition of Gaussian noise. The marginal at diffusion step t satisfies

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}),$$

where $\bar{\alpha}_t \in (0, 1)$ is the noise schedule. When p_{data} is Gaussian, the marginal $q(\mathbf{x}_t)$ is a Gaussian convolution of the noise kernel with the data distribution, and Theorem 5.1 supplies the exact reverse step.

8.3 Gaussian Process Regression

A Gaussian process (GP) prior $f \sim \mathcal{GP}(0, k)$ combined with i.i.d. observation noise $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$ yields noisy measurements $y_i = f(\mathbf{x}_i) + \varepsilon_i$ [8]. The predictive distribution of a new observation y_* at test point $\mathbf{x}_*$ is Gaussian with variance $\sigma_{y_*}^2$. The distribution of the underlying latent value f_* is obtained by Gaussian decimation: removing the noise variance σ_n^2 from the predictive variance, giving a latent-function posterior with variance $\sigma_{y_*}^2 - \sigma_n^2$.

8.4 Summary of AI/ML Connections

Table 3 collects the four leading examples. In each case the “known component” m is a Gaussian noise or prior kernel, the “composite” n is an observable or predicted Gaussian marginal, and the “recovered d ” is the latent signal or data distribution obtained by decimation.

Table 3: Gaussian decimation arising in representative AI and ML models.

Model	Known component m	Composite n	Recovered d
VAE [6]	Prior $\mathcal{N}(\mathbf{0}, \mathbf{I})$	Marginal $p_\theta(\mathbf{x})$	Decoder noise covariance
Diffusion [4]	Noise kernel $\mathcal{N}(\mathbf{0}, (1-\bar{\alpha})\mathbf{I})$	Noisy marginal $q(\mathbf{x}_t)$	Data dist. p_{data}
GP regression [8]	Obs. noise $\mathcal{N}(0, \sigma_n^2)$	Predictive $p(y_*)$	Latent posterior f_*
Kalman filter [5]	Transition noise	Predicted state density	Filtered state density

9 Geometric Visualisation

Figure 1 illustrates the three densities for the concrete example $m \sim \mathcal{N}(0, 1)$, $n \sim \mathcal{N}(2, 4)$, yielding the residual $d \sim \mathcal{N}(2, 3)$.

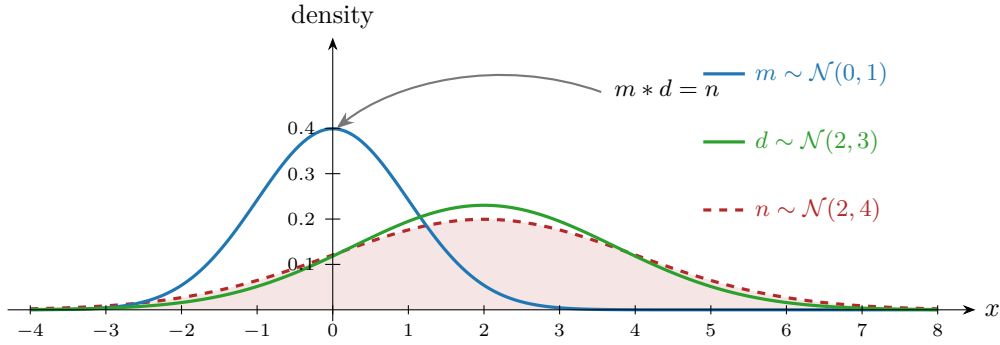


Figure 1: Gaussian decimation with $m \sim \mathcal{N}(0, 1)$ and $n \sim \mathcal{N}(2, 4)$. The residual density $d \sim \mathcal{N}(2, 3)$ (green) is the unique Gaussian that, when convolved with m (blue), reproduces the composite n (red, dashed). The shaded region indicates the spread of n .

Figure 2 gives a complementary bar chart of the additive variance decomposition $\tau^2 = \sigma^2 + (\tau^2 - \sigma^2)$ that underlies Theorem 3.1, using the same numerical example.

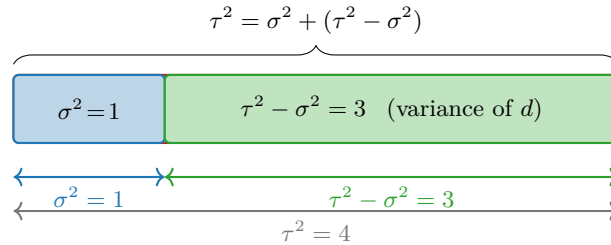


Figure 2: Variance decomposition for $\sigma^2 = 1$ and $\tau^2 = 4$. The total variance τ^2 of the composite n partitions additively into the variance σ^2 of the known component m (blue) and the residual variance $\tau^2 - \sigma^2$ of the decimating density d (green). Each width is proportional to the corresponding variance.

10 Conclusion

We have characterised the unique Gaussian density d that, when convolved with a known Gaussian $m \sim \mathcal{N}(\mu, \sigma^2)$, reproduces a target Gaussian $n \sim \mathcal{N}(\nu, \tau^2)$. The solution $d \sim \mathcal{N}(\nu - \mu, \tau^2 - \sigma^2)$

follows from a single application of the convolution–multiplication duality (Lemma 2.4) and the Lévy uniqueness theorem. The result is sharp: a valid Gaussian solution exists if and only if $\tau^2 > \sigma^2$, and degenerates gracefully to a Dirac delta at the boundary $\tau^2 = \sigma^2$, $\nu = \mu$. Theorem 5.1 extends the result to p dimensions via positive-definiteness of the residual covariance matrix.

Beyond its mathematical elegance, Gaussian decimation is a foundational operation in modern probabilistic computing. It underlies the prediction–update cycle of the Kalman filter, the encoder–decoder decomposition of variational autoencoders, the forward–reverse analysis of score-based diffusion models, and the noise–signal separation in Gaussian process regression. The unified characteristic-function framework presented here clarifies the common algebraic structure behind these otherwise disparate applications and suggests further generalisations—for instance, to the broader exponential family or to non-commutative convolutions on Lie groups.

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Glossary

Characteristic function

The Fourier transform of a probability density: $\hat{f}(\xi) = \mathbb{E}[e^{i\xi X}]$, $\xi \in \mathbb{R}$. Converts convolution of densities into pointwise multiplication (Lemma 2.4), which is the key analytic tool in the paper.

Convolution ($f * g$)

An integral operation on two functions: $(f * g)(x) = \int_{-\infty}^{\infty} f(x-t) g(t) dt$. If $X \sim f$ and $Y \sim g$ are independent, then $X + Y \sim f * g$.

Dirac delta (δ)

A generalised function satisfying $\int_{-\infty}^{\infty} \delta(x) f(x) dx = f(0)$ for every continuous f . Acts as the identity element of convolution: $f * \delta = f$ for every density f .

Gaussian (normal) distribution $\mathcal{N}(\mu, \sigma^2)$

A continuous distribution on $\mathbb{R}$ fully characterised by its mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$, with bell-shaped density $\phi_{\mu, \sigma^2}(x) \propto \exp(-(x - \mu)^2 / (2\sigma^2))$.

Gaussian decimation

The operation of finding the unique density d such that $m * d = n$, given two Gaussian densities m and n . A valid Gaussian solution exists if and only if the target n is strictly more spread than the component m (i.e., $\tau^2 > \sigma^2$).

Gaussian process (GP)

A stochastic process in which every finite collection of function values is jointly Gaussian, characterised by a mean function and a covariance (kernel) function k .

Kalman filter

A recursive Bayesian estimator for linear Gaussian state-space models, alternating between a prediction step (Gaussian convolution with the process-noise kernel) and an update step (Bayesian conditioning on the observation). See [5].

Lévy uniqueness theorem

The theorem guaranteeing that a probability distribution on $\mathbb{R}^p$ is uniquely determined by its characteristic function. It justifies identifying d from $\hat{d}$ in Steps 3 and 4 of the proof of Theorem 3.1.

Positive-definite matrix ($\mathbf{A} \succ 0$)

A symmetric real matrix satisfying $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all non-zero $\mathbf{x} \in \mathbb{R}^p$. The multivariate analogue of a positive scalar variance; $\mathbf{T} - \mathbf{\Sigma} \succ 0$ is the necessary and sufficient condition for multivariate Gaussian decimation (Theorem 5.1).

Stability (Gaussian)

The closure property: if $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then $X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Gaussian decimation is the exact converse of this property.

Variance

A measure of dispersion: $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$. Additive over independent random variables ($\text{Var}(X + Z) = \text{Var}(X) + \text{Var}(Z)$); the fundamental scalar quantity controlling when Gaussian decimation is feasible.

Variational autoencoder (VAE)

A latent-variable generative model [6] that learns a probabilistic encoder $q_\phi(\mathbf{z} \mid \mathbf{x})$ and decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$ by maximising an evidence lower bound (ELBO), with a standard Gaussian prior on the latent space.

The End